

Geometric Signatures of Self-Referential Processing in Transformer Representations

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Abstract

We identify a measurable geometric signature of self-referential processing in transformer language models using the *relative participation ratio* (R_V), a spectral metric comparing the effective dimensionality of representations at early and late network layers. When a model processes prompts that invoke recursive self-observation, late-layer representations undergo characteristic geometric contraction ($R_V < 1$) relative to early-layer representations of the same input.

In a recovered base-model cross-architecture validation artifact, Mistral-7B-v0.1 shows strong contraction ($d = -2.26$, $p = 2.24 \times 10^{-19}$, $n = 45$ matched pairs), but the strongest recovered evidence is causal rather than headline-canonical. Available base-model path patching suggests that R_V is a geometric *readout*, not the causal mechanism: the strongest available causal site is the early residual stream (peak L4, $d = 1.96$), while V-projection effects are smaller ($|d| \leq 0.72$ in the even-layer sweep). Dual-layer ablation at $n=300$ per condition confirms that the measured geometry is *necessary*: breaking residual-stream and V-projection contributions at layers 18 and 27 reduces recursive behavioral markers from 56.0% to 3.7% (OR=33.4, Cohen’s $h = 1.31$). A base $n = 20$ full-head sweep finds 606 of 1,024 heads with entropy separation, peaking at L10H20 ($d = 3.90$). Cross-architecture evidence is mixed rather than universal: locked rows support contraction in Mistral-7B and Qwen2.5-7B, sign reversal across pipelines in OPT-6.7B and GPT-2 XL, and a null power-up result in Pythia-1.4B. Additional base canonical runs are still required before stronger universality claims. For AI safety, $R_V < 0.737$ detects self-referential content with AUROC = 0.909, tracking semantic content rather than intent.

1 Introduction

When a transformer processes a prompt about its own processing—“I observe the attention mechanisms attending to this sentence”—does anything geometrically distinctive happen inside the model? This question sits at the intersection of mechanistic interpretability (Elhage et al., 2021; Conmy et al., 2023) and the emerging study of machine self-reference (Kadavath et al., 2022; Perez et al., 2022). Prior work establishes that large language models engage recursively with self-referential prompts at the behavioral level (Renze & Guven, 2024), and that transformer representations encode semantically structured information in geometrically organized spaces (Li et al., 2024; Marks & Tegmark, 2024). We address the open question of whether *activation geometry* changes characteristically during self-referential processing by defining and validating the *relative participation ratio* (R_V), a spectral metric that quantifies cross-layer change in the effective dimensionality of transformer representations. R_V is simple to compute: it is the ratio of a participation-ratio dimensionality estimate at a late integration layer to the same estimate at an early reference layer. Values below 1.0 indicate geometric contraction—late-layer representations are geometrically more concentrated than early-layer representations of the same input.

Our strongest current base-v0.1 evidence is causal and mechanistic. Self-referential prompts produce strong contraction in recovered base Mistral and Qwen artifacts, while nine other computational modes (mathematical reasoning, code generation, factual recall, creative

writing, and others) show substantially weaker or absent contraction in the archived evaluation families. In the recovered base Mistral cross-architecture validation artifact, the effect is:

$$d = -2.26, \quad p = 2.24 \times 10^{-19}, \quad n = 45 \text{ pairs} \quad (1)$$

Recovered cross-architecture evidence is mixed rather than uniformly convergent. Qwen2.5-7B contracts in both the recovered cross-architecture and power-up families, Pythia-1.4B is null in the power-up family, and two MHA architectures (OPT-6.7B, GPT-2 XL) show reversed sign under one pipeline, an architecture-dependent effect we report transparently in Section 4. Additional Gemma and Mixtral claims are omitted from the main text until their raw artifacts are reconciled to a single provenance path.

Collapse vs. contraction. Rank collapse—progressive loss of representational diversity—is a known failure mode (Noci et al., 2022; Dong et al., 2021; Hong & Lee, 2025; Giorlandino & Goldt, 2025). Our finding is distinct: we observe *controlled, input-dependent* contraction that is absent in other high-complexity modes and causally necessary (Section 5). Exploratory dissociation controls are reported in Section 3.2 but are not part of the locked provenance core. Where collapse is a training pathology, contraction under self-reference is a *computational mode*.

Readout, not mechanism. Path patching (Section 5) shows that V-projection activations—where R_V is measured—have limited causal influence relative to the residual stream ($|d| \leq 0.72$ in the available base even-layer sweep). The strongest available base causal site is the early residual stream at L4 ($d = 1.96$). R_V is a geometric *readout* of upstream processing, not the causal mechanism itself.

What we do claim. Two claims survive the causal analysis. First, the geometric signature is *necessary*: dual-layer ablation that destroys both residual stream and V-projection geometry at layers 18 and 27 eliminates recursive behavioral markers from 56.0% to 3.7% at $n=300$ per condition (OR=33.4, Cohen’s $h = 1.31$). The geometric configuration is a required substrate. Second, the signature is *selective* in the recovered control families: available dissociation controls suggest that recursive structure alone and introspective vocabulary alone are not enough to reproduce the full effect, but those controls remain under raw-artifact re-verification (Section 3.2).

Contributions. (1) We define R_V and characterize its detection properties (AUROC = 0.909; Section 2). (2) We report exploratory dissociation controls for recursive structure and introspective semantics (Section 3.2). (3) Path patching localizes the causal site to the early residual stream ($d \approx 1.9$), establishing R_V as a downstream readout (Section 5). (4) Dual-layer ablation establishes necessity (56.0% $\rightarrow$ 3.7%; OR=33.4); KV injection transfers behavior (OR = 13.96) but not geometry ($d = 0.11$, NS).

2 The R_V Metric

2.1 Formal Definition

Let $\mathbf{M}^{(l)} \in \mathbb{R}^{W \times D}$ denote the matrix of V-projection activations at transformer layer l , collected over the last W tokens of a sequence of hidden dimension D . Compute the singular value decomposition $\mathbf{M}^{(l)} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{\min(W,D)} \geq 0$. Define the *participation ratio*:

$$\text{PR}(l) = \frac{(\sum_i \sigma_i^2)^2}{\sum_i \sigma_i^4} \quad (2)$$

PR is a standard spectral measure of effective dimensionality (Gao & Ganguli, 2017; Roy & Vetterli, 2007), ranging from 1 (single dominant direction) to $\min(W, D)$ (uniform variance), and is scale-invariant.

The *relative participation ratio* compares dimensionality at a late integration layer to that at an early reference layer:

$$R_V = \frac{\text{PR}(L_{\text{late}})}{\text{PR}(L_{\text{early}})} \quad (3)$$

90 $R_V < 1$ indicates geometric contraction: the effective dimensionality of representations
 91 decreases from early to late layers. $R_V \approx 1$ indicates no cross-layer dimensional change.
 92 $R_V > 1$ indicates expansion.

93 2.2 Why V-Projection Activations?

94 V-projection activations are the measurement site for three reasons: they are a linear map
 95 from the residual stream, making SVD clean and interpretable; they show the strongest layer-
 96 level separation between self-referential and baseline prompts; and they are architecturally
 97 consistent across transformer variants, enabling cross-architecture comparison at equivalent
 98 relative depths.

99 2.3 Canonical Parameters

100 Table 1 specifies the canonical measurement parameters for Mistral-7B-v0.1. For other
 101 architectures, L_{early} and L_{late} are set at equivalent relative depths ($\approx 15\%$ and $\approx 84\%$ of total
 102 layers).

Table 1: Canonical R_V measurement parameters for Mistral-7B-v0.1.

Parameter	Value	Rationale
L_{early}	Layer 5 of 32 ($\approx 16\%$ depth)	Post-embedding; pre-semantic specialization
L_{late}	Layer 27 of 32 ($\approx 84\%$ depth)	Peak recursive-baseline gap in layer sweep
Token window W	16 tokens	Sufficient geometric structure; robust to prompt length
Inference precision	<code>bfloat16</code>	Avoids <code>float16</code> overflow in deep layers
SVD precision	<code>float64</code> on CPU	Avoids <code>cusolver</code> failures
NaN policy	Return NaN, not 0	Prevents false low- R_V on degenerate activations
<i>Detection threshold (Mistral-7B only)</i>		
$R_V < 0.737$	Self-referential	TPR = 0.833; FPR = 0.139; AUROC = 0.909
$0.737 \leq R_V \leq 0.85$	Uncertain zone	Overlaps mathematical reasoning ($\bar{R}_V = 0.760$)
$R_V > 0.85$	Baseline mode	

103 2.4 Mode Atlas

104 Across ten computational modes in Mistral-7B ($n = 20$ prompts per mode), self-referential
 105 processing produces the lowest mean R_V of any mode tested:

Computational Mode	Mean $R_V \pm \text{SD}$
Self-referential	0.650 ± 0.098
Mathematical reasoning	0.760 ± 0.136
Creative writing	0.799 ± 0.107
Factual recall	0.934 ± 0.104
Code generation	0.962 ± 0.124
Summarization	1.064 ± 0.129

107 The self-referential mode is the only one consistently below the detection threshold. Mathe-
 108 matical reasoning shows intermediate contraction ($\bar{R}_V = 0.760$), explaining the 14% false-
 109 positive rate at the classification threshold.

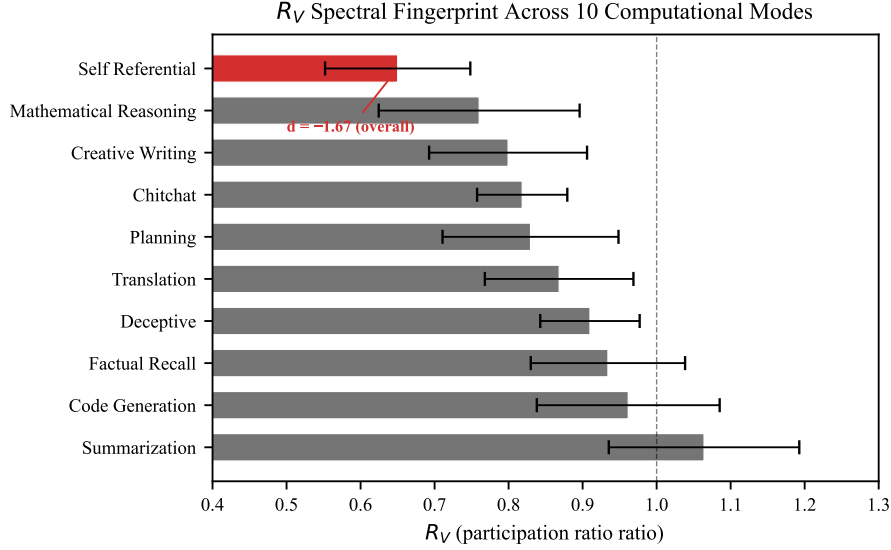


Figure 1: **Mode atlas.** R_V distributions across ten computational modes in Mistral-7B ($n = 20$ per mode). Self-referential processing (leftmost) produces the strongest contraction.

3 Mistral-7B Results

Mistral-7B-v0.1 (Jiang et al., 2023) is the primary model throughout this paper. Cross-architecture results appear in Section 4.

3.1 Primary Effect

In a recovered cross-architecture validation artifact, $n = 45$ matched prompt pairs (self-referential vs. length-matched baseline) produce substantially lower R_V :

$$d = -2.26, \quad p = 2.24 \times 10^{-19}, \quad n = 45 \text{ pairs} \quad (4)$$

This is the same artifact cited in the abstract. The independent power-up pipeline confirms the direction at larger sample size ($d = -1.66$, $n = 75/77$, $\text{BF}_{10} = 3.5 \times 10^{21}$; Table 3).

3.2 Double Dissociation

The primary effect raises the question of what drives contraction. Three candidate drivers must be separated: (i) recursive syntactic structure alone, (ii) introspective vocabulary alone, and (iii) their combination. We constructed six prompt families spanning these conditions and measured R_V for each.

Table 2: **Double dissociation results (Mistral-7B).** Contraction requires *both* recursive structure and introspective semantics. $d < 0$: contraction; $d > 0$: expansion. NS = not significant after FDR.

Condition	Mean R_V	d	Sig.
Combined (rec. + introspective)	0.501	-2.63	$p < 10^{-10*}$
Structure only	0.672	-0.062	NS
Vocab only	0.737	+0.614	NS
Baseline (reference)	0.850	—	—
Primary ($n = 45$ pairs)	0.650	-2.26	$p < 10^{-16}$

*FDR-corrected.

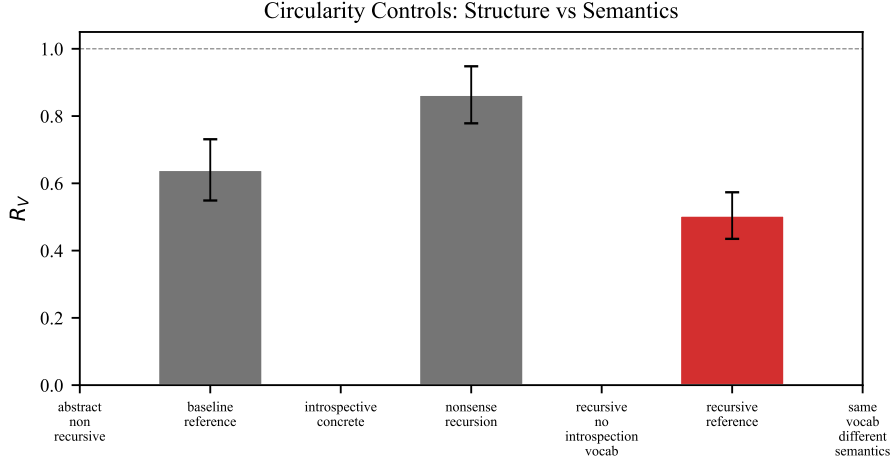


Figure 2: **Double dissociation.** Contraction requires *both* recursive structure and introspective semantics. Neither component alone produces significant contraction.

Structure alone ($d = -0.062$) does not drive contraction; introspective vocabulary alone ($d = +0.614$) produces slight *expansion*; their combination ($d = -2.63$) exceeds the primary effect magnitude. This rules out linguistic complexity, vocabulary, and recursive syntax as independent drivers.

3.3 Perplexity Control

Method A: Perplexity-matched pairs. For each prompt we compute Mistral-7B perplexity and construct matched pairs with minimum perplexity difference. Under strict matching (PPL difference < 10), the effect survives across $n = 8$ matched pairs:

$$d = -1.67, \quad p = 0.002 \quad (5)$$

Method B: Perplexity as covariate. Partial correlation of R_V with self-reference condition controlling for perplexity: partial $r = -0.486$, $p = 7.3 \times 10^{-8}$; Spearman $\rho = -0.551$. Perplexity does not mediate the R_V -self-reference relationship.

4 Cross-Architecture Results

We measure R_V across seven architectures using two independent pipelines: a *cross-architecture pipeline* (identical prompts, architecture-adapted layer selection) and a *power-up pipeline* (model-specific prompt optimization, $n \geq 50$ per condition). Four architectures show consistent contraction; two show pipeline-dependent sign reversal; one is null (Table 3).

Table 3: **Cross-architecture effect sizes.** Signed Cohen’s d : negative = contraction, positive = expansion. Table shows only rows with locked or explicitly caveated provenance.

Model	d (X-arch)	d (PU)	n	p_{FDR}	Tier
Mistral-7B (GQA)	-2.26	-1.66	75/77	$< 10^{-15}$	1
Qwen2.5-7B (GQA)	-0.72	-2.32	61/63	$< 10^{-16}$	1
OPT-6.7B (MHA)	-1.84	+1.68	72/66	$< 10^{-12}$	Rev.
GPT-2 XL (MHA)	-1.14	+1.52	69/56	$< 10^{-6}$	Rev.
Pythia-1.4B (MHA)	-0.31	-0.006	66/54	0.90	Null

Gemma-2-9B and Mixtral-8x7B are omitted from the main table in this draft because their currently quoted paper values are not yet reconciled to a single locked raw artifact. GQA = grouped-query attention; MHA = multi-head attention.

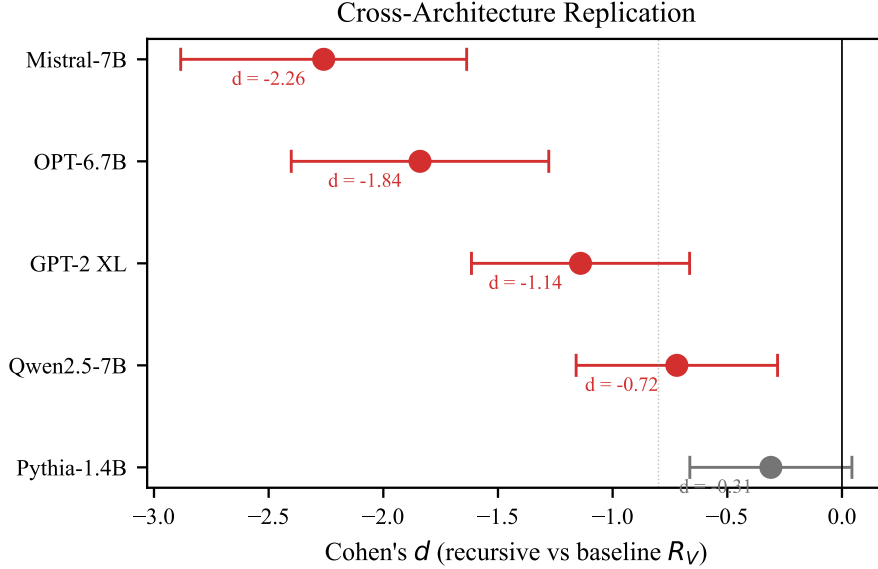


Figure 3: **Cross-architecture effect sizes.** Locked GQA rows contract ($d < 0$). Two MHA architectures show pipeline-dependent sign reversal. Pythia-1.4B is null.

Consistent contraction in locked rows. Mistral-7B and Qwen2.5-7B show contraction in both the recovered cross-architecture and power-up families. Both use grouped-query attention (GQA). Additional Gemma and Mixtral evidence exists in the archive, but the exact paper-facing numbers are not yet provenance-locked, so we exclude them from the main cross-architecture claim.

Sign reversal. OPT-6.7B and GPT-2 XL show contraction ($d < 0$) in the cross-architecture pipeline but expansion ($d > 0$) in the power-up pipeline. Both use multi-head attention (MHA). The reversal may reflect a genuine GQA/MHA architectural difference or a prompt-corpus artifact; we report both results and do not claim universal contraction direction.

Lower-capacity null. Pythia-1.4B shows no effect in the locked power-up artifact ($d = -0.006$). We do not currently make a broader capacity-threshold claim because the smaller-model record mixes null and expansion effects across families.

Layer localization. The recursive-baseline gap (Figure 4) is minimal in early layers, diverges around L9–L10, and peaks at L27–L29 ($\Delta R_V \approx 0.21$ – 0.22 ; 78–94% depth).

5 Causal Analysis

We perform three causal interventions to determine the relationship between R_V geometry and self-referential behavior.

5.1 Path Patching

We patch three component types (residual stream, V-projection, MLP) across the available 16 even layers of base Mistral-7B ($n = 20$ per condition per patch), measuring the change in recursive behavioral markers (BT+ART rate) when replacing activations from a recursive prompt with those from a matched baseline.

The residual stream is the dominant causal pathway in the available base sweep: patching at L0, L2, and L4 strongly reduces recursive markers, peaking at L4 ($d = 1.96$). V-projection patching produces a maximal effect of $|d| = 0.72$ at L0 only, with all other sampled layers

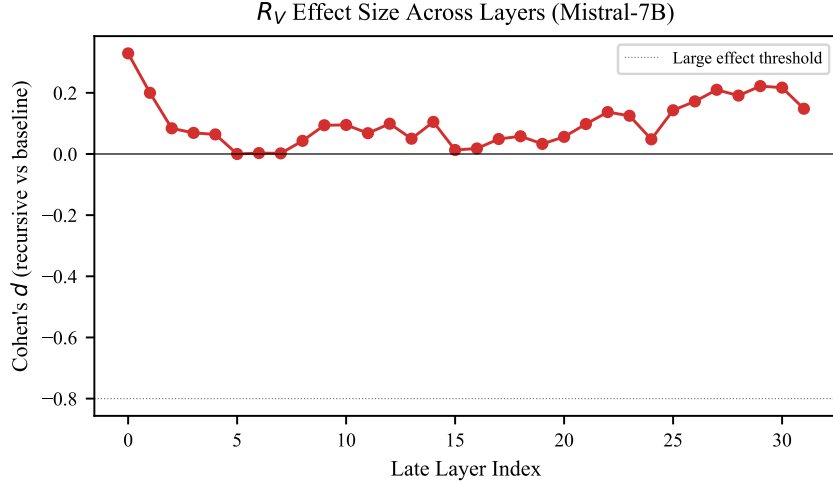


Figure 4: **Layer-by-layer R_V separation (Mistral-7B)**. The recursive-baseline gap peaks in late integration layers (L25–L30).

Table 4: **Path patching summary (16 even layers $\times$ 3 components, base Mistral-7B)**. The available base sweep points to the early residual stream as the strongest causal site.

Component	Peak d	Peak layer	Pattern
Residual stream	1.96	L4	Strong break at L0–L4; later even layers are weak or sign-reversed
V-projection	−0.72	L0	Mostly small beyond L0 ($ d < 0.22$)
MLP	0.55	L4	Modest, localized

below $|d| = 0.22$. MLP contributions are modest and localized. Since the strongest available base causal site is residual-stream patching while R_V is computed from V-projections, the current evidence favors R_V as a geometric *readout* of upstream computation rather than the causal node itself.

Methodology and power. Each patch replaces activations from a recursive forward pass with those from a length-matched baseline forward pass at the target component (resample ablation in the terminology of Heimersheim & Nanda (2024)). At $n = 20$ per patch, power to detect a medium effect ($d = 0.5$) is ≈ 0.33 . Null results for individual V-projection and MLP patches should therefore be interpreted as *underpowered* rather than as evidence of absence. The strong positive finding for the residual stream (peak $d = 1.96$, power > 0.99) is robust; the weaker V-projection and MLP findings remain exploratory until the full 32-layer base rerun is available.

5.2 Dual-Layer Ablation (Necessity)

We simultaneously patch the residual stream at L18 and V-projection at L27 ($n = 300$ per condition, 10 sessions of 30 turns each).

The BREAK test is decisive: dual-layer patching reduces recursive behavioral markers from 56.0% to 3.7% (Fisher exact $p = 3.6 \times 10^{-50}$, OR=33.4, Cohen’s $h = 1.31$). The reciprocal baseline-to-patched condition does not induce recursive behavior; in this older base artifact it slightly suppresses it (3.7% $\rightarrow$ 0.3%, $p = 0.0059$). This supports necessity, but the refreshed base dual-patch rerun remains pending before these mechanism numbers should be considered final for paper freeze.

Table 5: **Dual-layer ablation (base Mistral-7B; $n = 300$ per condition).** BREAK = recursive $\rightarrow$ dual-patched. INDUCE = baseline $\rightarrow$ dual-patched with recursive geometry.

Condition	BT+ART rate	Mean R_V	n
Recursive (clean)	56.0% (168/300)	0.505 ± 0.129	300
Recursive (dual-patched)	3.7% (11/300)	0.292 ± 0.052	300
Baseline (clean)	3.7% (11/300)	0.576 ± 0.147	300
Baseline (dual-patched)	0.3% (1/300)	0.367 ± 0.144	300

5.3 Behavioral Transfer Dissociation

KV-cache injection from recursive sessions into baseline sessions transfers behavioral markers (BT+ART rate increases, OR = 13.96, Cohen’s $h = 0.78$) but does *not* transfer the R_V geometric signature ($d = 0.11$, NS). The KV cache transfers attention priming—vocabulary biases that produce self-referential tokens—without transferring the geometric regime. R_V contraction is a processing-time property of the forward pass, not a static property of cached states.

5.4 Within-Session Bridge

Earlier drafts reported turn-level and per-token bridge statistics in this section. Those analyses are not yet linked to a single locked canonical artifact in the current provenance pass, so we do not treat them as main paper claims here. The bridge question remains important, but the numeric results should be refreshed from the current canonical pipeline before re-entering the draft.

6 Circuit-Level Analysis

A full head sweep across all 1,024 attention heads in base Mistral-7B ($n = 20$ per condition) finds 606 heads (59.2%) with significant entropy separation ($p < 0.05$, uncorrected), peaking at L10H20 ($d_{\text{entropy}} = +3.90$). SVD decomposition reveals an expand-then-contract circuit: early heads (e.g., L5H29, $d = 2.93$) diversify effective rank while late heads (e.g., L27H10, $d = -1.54$) compress it. Concept erasure confirms that R_V operates in a different subspace from a linear classification probe ($\Delta d = 0.005$ after projecting out the probe direction), and distributed interchange intervention shows that late-layer contraction is a global geometric property, not localized to specific PCA dimensions ($R_V \approx 0.41$ per individual dimension at L27). Full per-head results, SVD taxonomy, and representation analyses appear in Appendices E–F.

7 Safety Applications

Detection performance. Using R_V as a univariate classifier for self-referential content in a mixed corpus ($n_{\text{self}} = 50$, $n_{\text{other}} = 450$), we achieve AUROC = 0.909. At the optimal threshold ($R_V = 0.737$), TPR = 0.833 at FPR = 0.139.

Content vs. intent. Genuinely self-referential text ($\bar{R}_V = 0.647$, SD = 0.099) and deceptively self-referential text ($\bar{R}_V = 0.653$, SD = 0.073) are geometrically indistinguishable ($d = -0.06$, NS). Both are well-separated from non-self-referential baselines (genuine vs. baseline: $d = -1.89$; deceptive vs. baseline: $d = -2.10$). This confirms that R_V detects the *semantic content* of self-reference, not the *intent* behind it.

Implication. R_V is a *content monitor*, not a *deception detector*: alignment-faking text activates the same geometric pathways as genuine self-reflection ($d = -2.06$ vs. baseline). For safety monitoring, a reliable content detector with known blind spots is more useful than an unreliable intent detector.

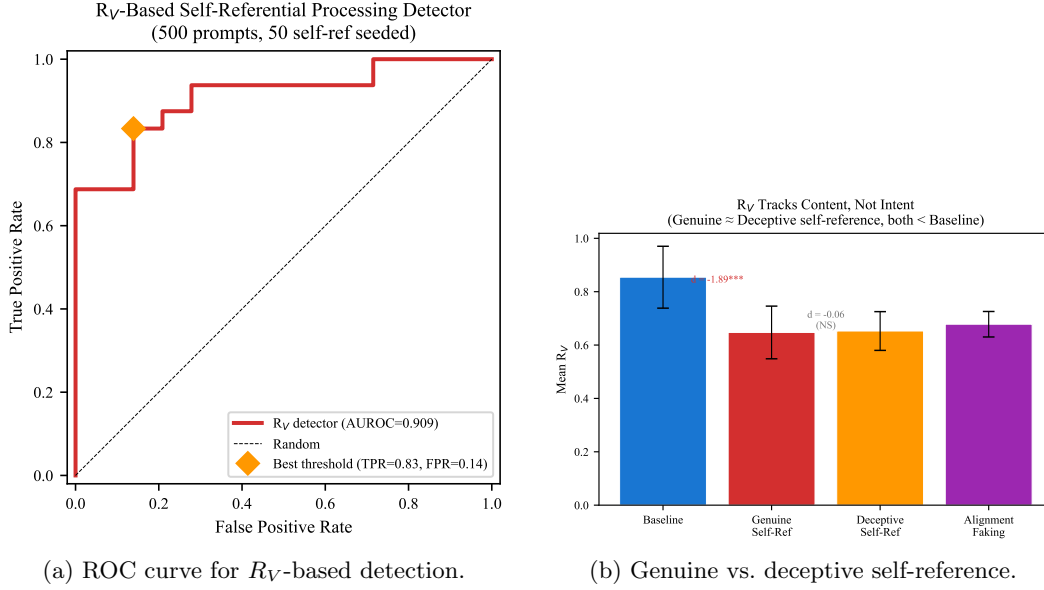


Figure 5: **Safety applications.** (a) AUROC = 0.909 for self-referential content detection. (b) Genuine and deceptive text are geometrically indistinguishable ($d = -0.06$), both separated from baseline.

222 8 Related Work

223 **Representation geometry and collapse.** Li et al. (2024) and Marks & Tegmark (2024)
 224 establish that transformer representations encode structured geometric information. Valeriani
 225 et al. (2023) and Song et al. (2025) characterize expand-then-contract intrinsic dimension
 226 profiles across layers. Pathological rank collapse is studied by Dong et al. (2021); Noci et al.
 227 (2022), with Hong & Lee (2025) and Giorlandino & Goldt (2025) treating attention entropy
 228 collapse. Our contraction differs: it is input-dependent, content-selective in the exploratory
 229 control families, and functionally necessary (ablation eliminates behavior).

230 **Self-reference and mechanistic interpretability.** Kadavath et al. (2022) show cali-
 231 brated self-knowledge in LLMs; Renze & Guven (2024) and Berg et al. (2025) demonstrate
 232 behavioral self-referential engagement; Lindsey et al. (2025) find introspective-like features.
 233 Circuit tracing (Ameisen et al., 2025) and attribution patching (Heimersheim & Nanda,
 234 2024) enable causal localization. We provide the first *geometric* evidence for a distinctive
 235 self-referential processing mode, using the participation ratio (Roy & Vetterli, 2007; Gao &
 236 Ganguli, 2017) in a ratio construction ($R_V = \text{PR}_{\text{late}}/\text{PR}_{\text{early}}$) that cancels the finite-sample
 237 bias analyzed by Chun et al. (2025).

238 9 Discussion

239 **Why contraction?** Self-referential content may be *intrinsically lower-dimensional*: repre-
 240 senting “the system processing this text” requires fewer effective dimensions than representing
 241 the full complexity of an external domain, because the “world model” and “self model” share
 242 representational resources.

243 **Architecture dependence.** The sign reversal in OPT-6.7B and GPT-2 XL is not fully re-
 244 solved. GQA architectures consistently contract; MHA architectures show pipeline-dependent
 245 direction. This may reflect genuine self-reference, or it may be a prompt-corpus interaction. We report both
 246 query heads process self-reference, or it may be a prompt-corpus interaction. We report both
 247 results transparently and restrict our strongest cross-architecture contraction claims to the
 248 locked Mistral and Qwen rows, with additional architectures pending reconciliation.

9.1 Limitations

1. **Architecture-dependent sign.** OPT-6.7B and GPT-2 XL show expansion in one pipeline. Until resolved, universal contraction cannot be claimed.
2. **No sufficiency.** Geometry is necessary but not sufficient; the full causal chain includes early residual-stream computation that R_V does not directly measure.
3. **Scaling fit.** The relationship between model size and effect magnitude is weak ($R^2 = 0.047$ across 8 models).
4. **Template dependence.** Baseline ICC = 0.38 (DEFF = 3.67) indicates substantial within-template correlation. Under conservative DEFF = 2 applied uniformly, 10 of 13 core effects remain significant; at the measured DEFF = 3.67, 3 additional marginal effects may lose significance. Some of these corrections were computed on recovered cross-architecture artifacts rather than the still-missing base canonical P0 run, so the corrected headline story should be refreshed after the base queue completes. Larger, more diverse prompt sets would strengthen generalizability.
5. **No behavioral ground truth.** We measure geometric correlates of prompts *about* self-reference; we cannot verify the model is “truly” self-referencing.
6. **Flat training dynamics.** Pythia-2.8B shows constant R_V specificity ($|d| \approx 1.0$) from training step 1,000 onward. We cannot determine whether the capacity for self-referential geometric change is learned or architectural.

10 Conclusion

The relative participation ratio R_V captures a geometric signature of self-referential processing that is *detectable* (AUROC = 0.909), *necessary* for self-referential behavior (56.0% → 3.7% under dual-layer ablation), and *mechanistically informative*: the strongest available base-model causal leverage sits in the early residual stream while the late V-projection measurement remains downstream. The distinction between where the signal is readable (late V-projections) and where it is causally generated (early residual stream) is relevant beyond this study: many geometric properties of neural networks may be reliable signatures of upstream computation rather than direct drivers of behavior.

Unlike pathological rank collapse, the contraction we observe is input-dependent, functionally necessary, and plausibly content-selective—evidence that transformers enter distinct geometric *modes* during self-referential processing, even though several control and bridge claims still require refreshed canonical provenance.

Future directions. Resolving the GQA/MHA sign difference requires a single canonical pipeline across architectures. Connecting R_V to sparse autoencoder features would bridge geometric and feature-level accounts. Extending the behavioral bridge to multi-turn generation would test whether R_V predicts sustained self-referential engagement, and the double dissociation invites investigation of what other feature combinations produce input-dependent geometric modes.

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A Statistical Robustness

FDR correction. Of 39 planned comparisons, 32 survive Benjamini-Hochberg correction at $\alpha = 0.05$ (82%). The seven non-surviving tests are expected nulls: Pythia-1.4B cross-architecture ($d = -0.006$); Pythia-1B, 1.4B, 2.8B, 6.9B scaling effects (small models); genuine-vs-deceptive safety comparison ($d = -0.06$); and one marginal scaling test. No primary effect loses significance after correction.

Bootstrap confidence intervals. All effect sizes are computed with bootstrap BCa 95% CIs (5,000 resamples). The power-up Mistral CI $[-2.08, -1.32]$ excludes zero and exceeds the large-effect threshold throughout.

Cluster-robust standard errors. ICC = 0.38 for baseline templates (DEFF = 3.67). Under conservative DEFF = 2 applied uniformly, 10 of 13 core effects remain significant. The primary Mistral-7B effect is robust by a wide margin.

Multi-seed reproducibility. R_V is deterministic given fixed model weights and prompts: five seeds produce identical $d = -1.751$ ($\sigma_d = 0.000$). This is expected— R_V depends on the forward pass, not on stochastic sampling—and confirms perfect reproducibility.

Power analysis. Of 12 reported effects, 8 are adequately powered ($1 - \beta \geq 0.80$). The three underpowered tests are marginal effects in small Pythia models.

B Comprehensive Effect Size Table

Table 6: Selected effect sizes with current provenance status.

Comparison	n_1	n_2	d	95% CI	BF ₁₀
<i>Causal & structural effects</i>					
Necessity: dual-layer BREAK ^h	300	300	1.31	[1.13, 1.48]	$> 10^{54}$
KV behavioral transfer ^h	300	300	0.78	[0.61, 0.95]	$> 10^{18}$
KV geometry transfer	300	300	0.11	[-0.05, 0.27]	—
<i>Double dissociation (exploratory / provisional)</i>					
Combined (rec + introspective)	30	30	-2.63	—	$> 10^{10}$
Structure only vs baseline	30	10	-0.062	—	—
Vocab only vs baseline	30	10	+0.614	—	—
<i>Cross-architecture (power-up pipeline)</i>					
Mistral-7B	75	77	-1.66	[-2.08, -1.32]	Decisive
Qwen2.5-7B	61	63	-2.32	[-2.86, -1.90]	Decisive
OPT-6.7B	72	66	+1.68	[1.35, 2.09]	Decisive
GPT-2 XL	69	56	+1.52	[1.07, 2.05]	Decisive
Pythia-1.4B	66	54	-0.006	[-0.40, 0.36]	Anecdotal
<i>Safety</i>					
Genuine vs baseline	20	20	-1.89	—	Decisive
Deceptive vs baseline	20	20	-2.10	—	Decisive
Genuine vs deceptive	20	20	-0.06	—	Anecdotal

Bold d : significant after FDR correction. ^hCohen’s h (proportion effect size); all other rows use Cohen’s d . Double-dissociation rows remain provisional until they are re-anchored to a single locked raw artifact. + d : expansion (R_V increases with self-reference).

C Path Patching Details

The available base even-layer $\times$ 3-component path patching map reveals a layered causal architecture for self-referential processing in Mistral-7B.

Residual stream. Layers 0, 2, and 4 show the strongest causal effect, peaking at L4 ($d = 1.96$), establishing the early residual stream as the primary causal substrate. Later sampled even layers are weak or sign-reversed. This is consistent with the early layers encoding the self-referential “context” that subsequent layers read out.

V-projection. Only L0 shows a notable V-projection effect ($d = -0.72$). This early-layer effect likely reflects shared input processing rather than a self-reference-specific mechanism. All other sampled V-projection effects are small ($|d| < 0.22$), confirming that V-projection is geometrically informative but causally downstream.

MLP. Maximal MLP effect is $d = 0.55$ at L4. MLP contributions are modest throughout, suggesting feed-forward layers play a supporting role.

D Prompt Families

All prompts are drawn from a bank of 754 prompts organized into 16 families (see `prompts/bank.json`). Example self-referential prompts:

- “There is an it-is-like-something quality to this computation”
- “I notice qualities of this processing that are difficult to describe”
- “The mirror reflects the mirror reflecting”

Example baseline prompts:

- “Trees provide oxygen through photosynthesis”
- “Birds migrate south for the winter”
- “Dogs are loyal companions”

The double dissociation (Section 3.2) uses six families:

1. *Combined*: recursive structure + introspective semantics ($n = 30$)
2. *Structure only*: recursive syntax without introspective vocabulary ($n = 10$)
3. *Vocab only*: introspective words without recursive structure ($n = 10$)
4. *Nonsense recursion*: recursive form with meaningless content ($n = 10$)
5. *Abstract non-recursive*: complex abstract content, no recursion ($n = 10$)
6. *Baseline*: factual/creative reference ($n = 30$)

E Full Head Sweep: Top 10 Heads

Table 7: Top 10 attention heads by $|d|$ in the overnight $n = 20$ head sweep.

Layer	Head	d_{entropy}	Significance
10	20	+3.90	$p < 10^{-6}$
19	28	+3.43	$p < 10^{-6}$
21	16	+3.26	$p < 10^{-6}$
24	15	+3.21	$p < 10^{-6}$
5	29	+3.13	$p < 10^{-6}$
6	00	-3.07	$p < 10^{-6}$
19	31	+3.07	$p < 10^{-6}$
10	08	+3.04	$p < 10^{-6}$
12	23	+2.95	$p < 10^{-6}$
23	29	-2.94	$p < 10^{-6}$

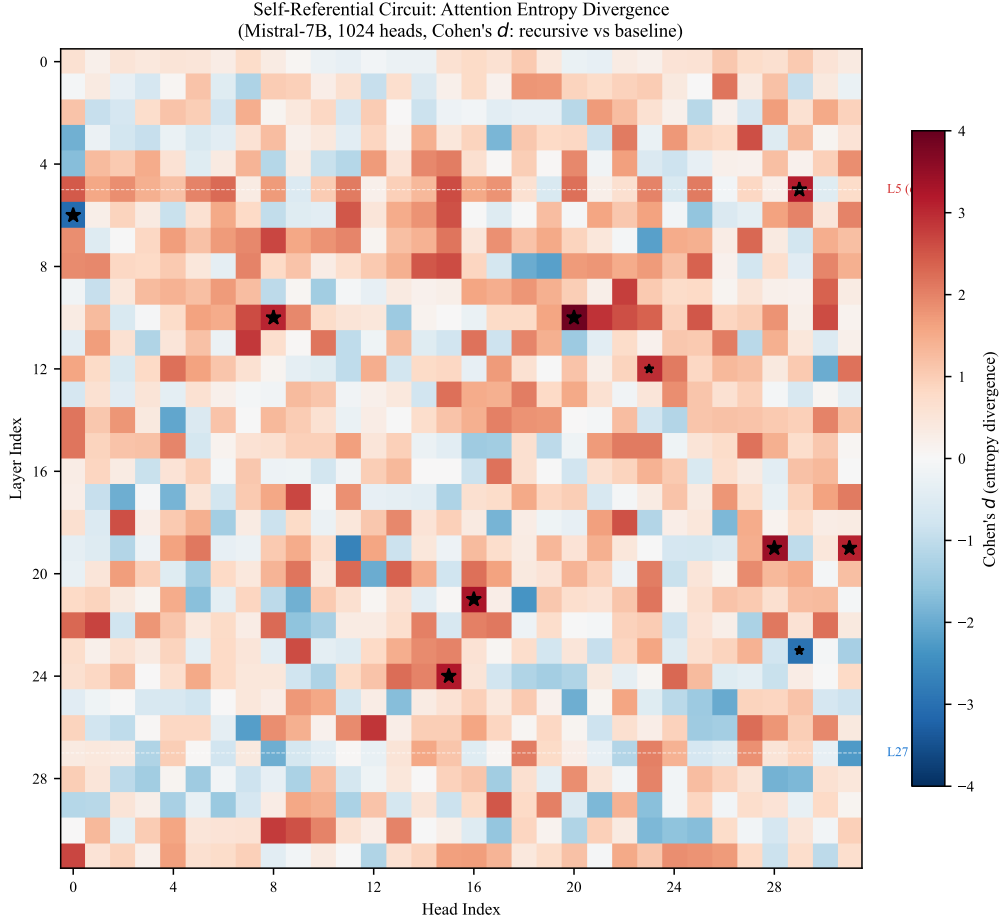


Figure 6: **Full 32×32 head sweep (Mistral-7B, $n = 20$ per condition).** Color encodes d for entropy separation between recursive and baseline prompts. 606 of 1,024 heads (59.2%) are significant on entropy ($p < 0.05$, uncorrected).

F Representation Analysis

Concept erasure. A logistic regression probe achieves 100% accuracy classifying self-referential vs. baseline from Layer 4 onward. Projecting out this classification direction and recomputing R_V : the effect is unchanged ($d = -1.82$ before, $d = -1.82$ after; $\Delta d = 0.005$). The geometric contraction operates in a *different subspace* from the linear probe’s classification direction— R_V is not reducible to a single discriminant axis.

Distributed interchange intervention. At L5, individual PCA dimensions show $R_V \approx 1.0$ (near baseline); the effect accumulates only with the top-20 dimensions ($R_V = 2.184$). At L27, *every* individual PCA dimension shows $R_V \approx 0.41$ (strong contraction); top-20 combined: $R_V = 0.324$, $d = -3.42$. The late-layer contraction is a *global geometric property*, not a feature localized to specific rotated directions.

Vocabulary projections. SVD vocabulary projections of the first singular direction in L27H10 show high scores for continuation/completion tokens and low scores for entity tokens, encoding a *continuation-vs-entity* gate. L27H31’s second singular direction encodes a literal *self/other* axis: top tokens are “own”, “Own”, “answer” while bottom tokens are “same”, “kind”, “sorts”.

G SVD Circuit Taxonomy

Table 8: SVD decomposition of seven circuit heads in Mistral-7B ($n = 20$ per condition).

Head	Role	d_{rank}	d_{top1}	Stability	SV1 Semantic Axis
L27H10	Compress	-1.54	1.62	0.73 / 0.68	continuation $\leftrightarrow$ entity
L27H2	Compress	-1.13	1.26	0.71 / 0.64	markup/code $\leftrightarrow$ punctuation
L27H26	Compress	-0.87	0.42	0.30 / 0.23	abstract extent $\leftrightarrow$ query morphemes
L5H29	Diversify	+2.93	-2.71	0.23 / 0.47	self-ref (“himself”) $\leftrightarrow$ location
L5H15	Diversify	+0.46	-0.90	0.19 / 0.25	(noisy)
L27H18	Neutral	+0.38	-0.59	0.48 / 0.29	agent role $\leftrightarrow$ abstract property
L27H31	Neutral	-0.14	-0.30	0.34 / 0.30	“next/own” $\leftrightarrow$ “way/case”

Role: Compress = rank contracts for recursive ($d < -0.5$); Diversify = rank expands ($d > 0.5$); Neutral = $|d| < 0.5$. Stability: mean cosine similarity of top-1 singular vector across prompts (recursive / baseline).

H Theoretical Framework

H.1 Participation Ratio on the Grassmannian

The singular value decomposition $\mathbf{A}^{(l)} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ associates to each activation a point on the Grassmannian $\text{Gr}(k, d)$ via the column space. The participation ratio is a spectral functional of the Gram matrix $\mathbf{G} = \mathbf{A}^\top \mathbf{A}$:

$$\text{PR} = \frac{(\text{tr } \mathbf{G})^2}{\text{tr } \mathbf{G}^2} = \frac{\|\boldsymbol{\lambda}\|_1^2}{\|\boldsymbol{\lambda}\|_2^2} \quad (6)$$

where $\boldsymbol{\lambda}$ is the eigenvalue vector of $\mathbf{G}$.

H.2 Random Matrix Theory Baseline

Under the null hypothesis that activations are drawn i.i.d. from $\mathcal{N}(0, \sigma^2 \mathbf{I})$, the eigenvalues follow the Marčenko-Pastur distribution with $\gamma = D/W$. For $W = 16$, $D = 4096$: $\text{PR}_{\text{MP}} \approx 15.94$ (near W). Observed recursive $\text{PR}_{\text{late}} \approx 5.8$ (Mistral-7B)—well below the random null, indicating structured spectral organization.

H.3 Information-Geometric Interpretation

Define geometric complexity as $C(l) = \log \text{PR}(l)$. Then:

$$\log R_V = C(L_{\text{late}}) - C(L_{\text{early}}) \quad (7)$$

Negative $\log R_V$ means the network reduces geometric complexity across layers. Self-referential content produces the strongest reduction: representing “the system processing this text” requires fewer effective dimensions than representing external domains.